

Cambridge (CIE) IGCSE Physics
Paper 3: Theory (Set E)
Core

Saturday 26 April 2025

Afternoon (Time: 1 hour 15 minutes)

Total marks

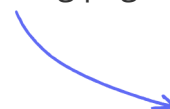
/ 80

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8N (acceleration of free fall = 9.8m/s^2).
- You should show all your working and use appropriate units.

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or visit the mock exams landing page for this course



- 1 (a)** A student has a piece of metal that has an irregular shape. The weight of the metal is 3.0 N.

Calculate the mass of the metal.

mass = kg

(2 marks)

- (b)** Fig. 2.1 shows the piece of metal, a measuring cylinder and a beaker containing water.

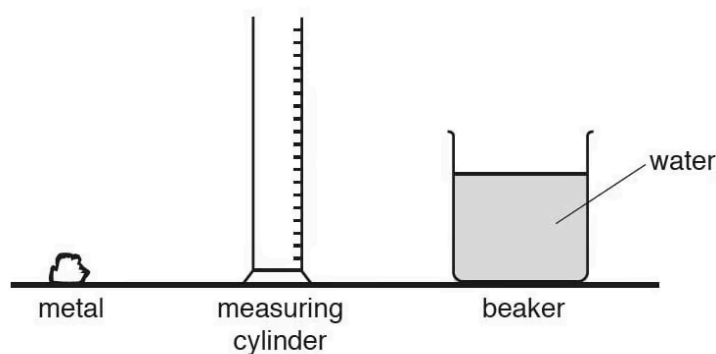


Fig. 2.1

- (i) Describe how to determine the volume of the metal, using the equipment in Fig. 2.1.

[4]

- (ii) Explain why the procedure in **(b)(i)** is not suitable for finding the volume of a piece of low-density wood that is of similar shape and size to the piece of metal in **(a)**.

[1]

- (iii) The mass of another piece of metal is 405 g and its volume is 150 cm³.

Calculate the density of the metal. State the unit.

density = [3]

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(8 marks)

- 2 (a)** Model trains move along a track passing through two model stations. Students analyse the motion of a train. They start a digital timer as the train starts to move. They record the time that it enters Station A and the time it enters Station B.

Fig 1.1 below shows the time on entering Station A and the time on entering Station B.



Fig. 1.1

Calculate the time taken from the train entering Station A to the train entering Station B.
State your answer in seconds.

time taken = s

.....
(1 mark)

- (b)** A faster train takes 54 s to travel from Station A to Station B. The distance between the stations is 120 m.

Calculate the average speed of this train.

average speed = m/s

.....
.....
.....

(3 marks)

(c) Fig. 1.2 shows the speed-time graph for a train travelling on a different part of the track.

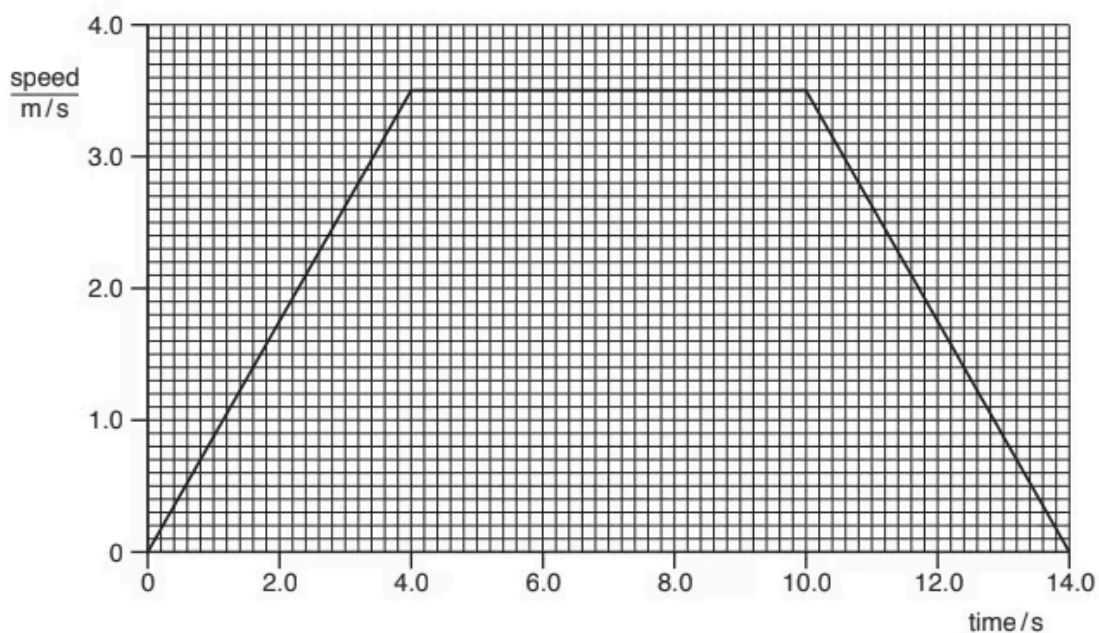


Fig. 1.2

Determine the total distance travelled by the train on this part of the track.

distance = m

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.....

.....

.....

(4 marks)

3 (a) Fig. 4.1 shows a pin. Fig. 4.2 shows a person pushing the pin into a wall.

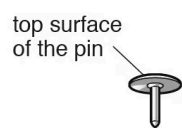


Fig. 4.1



Fig. 4.2

The area of the top surface of the pin is 1.8 cm^2 .

The person applies a force of 50 N.

Calculate the pressure exerted on the top surface of the pin.

pressure = N / cm^2

.....

.....

.....

(3 marks)

(b) The area of the top surface of the pin is 500 times larger than the area of the point.

Calculate the value of the pressure exerted by the point on the wall.

pressure = N / cm^2

.....

(1 mark)

- 4 (a)** A 50 cm rule is balanced at its mid-point. A force of 8.0 N acts at a distance of 10 cm from one end of the rule.

Fig. 2.1 shows the arrangement.

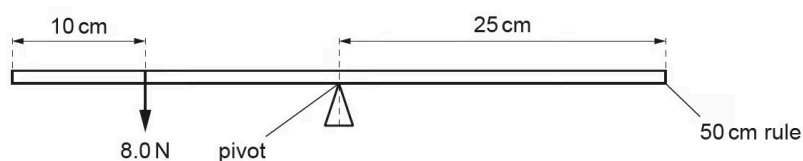


Fig. 2.1

Calculate the moment of the 8.0 N force about the pivot. Give the unit.

moment = unit =

.....

.....

.....

.....

.....

(5 marks)

- (b)** Another force acts at a point 10 cm from the pivot. It makes the rule balance.

On Fig. 2.1, draw an arrow to show the position and direction of this force.

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(2 marks)

5 (a) Fig.4.1 represents a hydroelectric system for generating electrical energy.

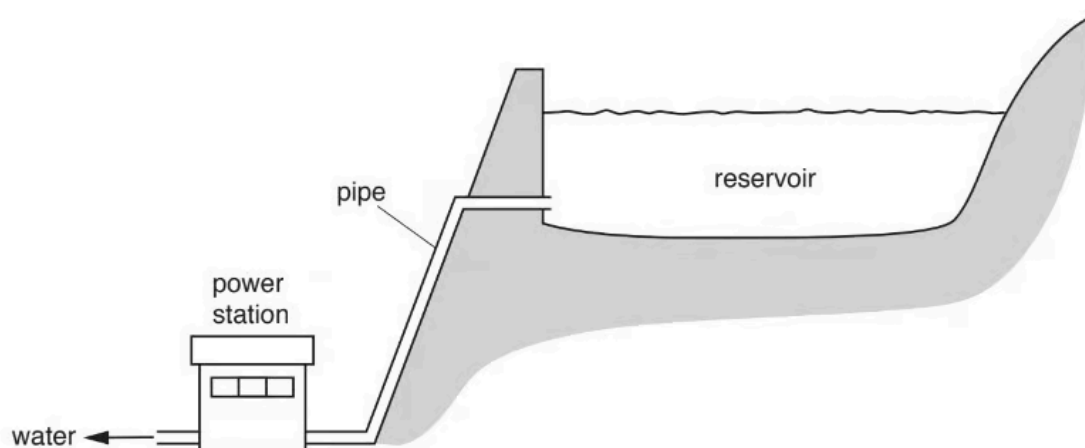


Fig.4.1

Describe how the power station uses the water in the reservoir to generate electricity.

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(3 marks)

(b) State three advantages of generating electricity using a hydroelectric system compared with using a coal-fired power station.

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(3 marks)

(c) An electric drill is shown in Fig.4.2.

Complete the sentences about the electric drill. Use words from the box. Each word may be used once, more than once, or not at all.

efficient	kinetic	potential	powerful	reliable	thermal
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(i) Energy is transferred electrically from the mains supply to the store of the drill.

[1]

(ii) A second electric drill transfers the same amount of energy per second as the first one. However, it wastes more of this energy. The second drill is less than the first drill.[1]

(2 marks)

6 (a) Fig. 5.1 shows a metal can containing air. The can is sealed with a lid.

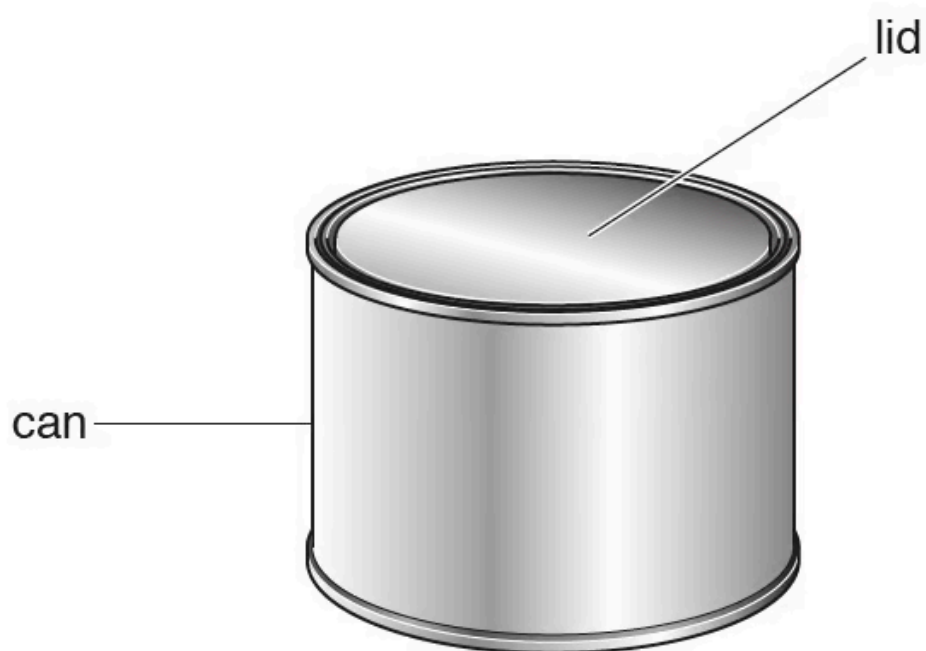


Fig. 5.1

The air in the can exerts a pressure of $20\,000\text{ N/m}^2$ on the lid. The area of the can lid is 0.09 m^2 .

Calculate the force on the lid due to the air in the can.

force = N

.....

.....

.....

(3 marks)

(b) The air in the can becomes warmer.

State and explain what happens to the pressure of the air in the can. Use your ideas about gas molecules.

(3 marks)

7 (a) Fig. 8.1 represents a travelling wave at an instant in time.

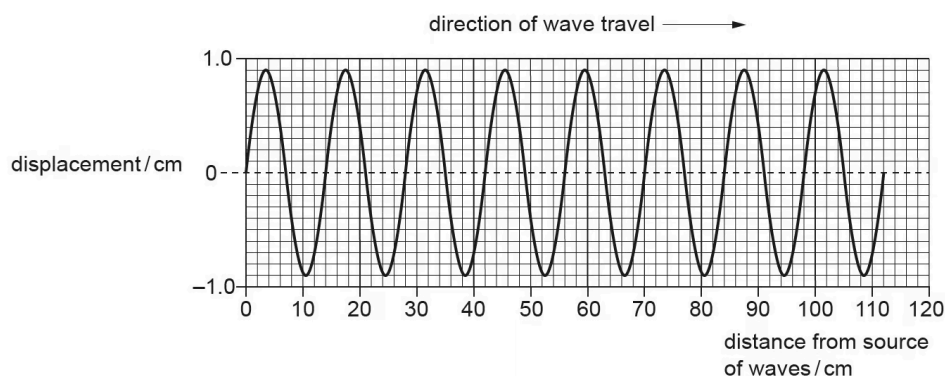


Fig. 8.1

(i) Determine the amplitude of the wave.

amplitude = cm [1]

(ii) Determine the wavelength of the wave.

wavelength = cm [2]

(iii) It takes 2.0 s for a source to emit the wave shown in Fig. 8.1.

Calculate the frequency of the wave.

frequency = Hz [2]

(5 marks)

(b) Fig. 8.2 shows the main regions of the electromagnetic spectrum.

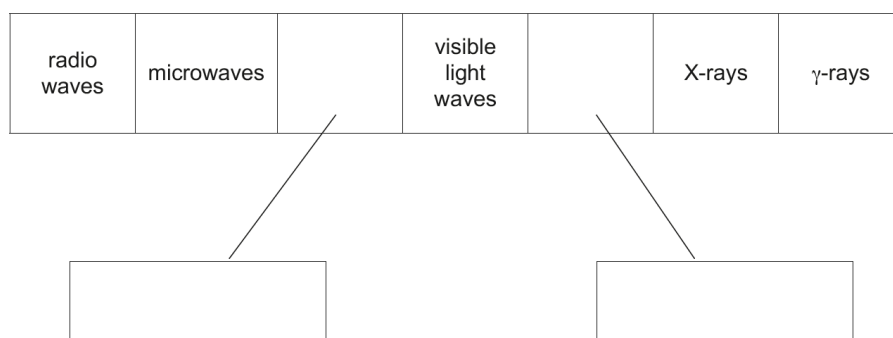


Fig. 8.2

(i) Two of the regions are not labelled.

Add the correct label to each of the unlabelled regions by writing in each box.

[2]

(ii) Describe one use of γ -rays.

[1]

(3 marks)

- 8 (a)** A wooden toy train has permanent magnets that hold the carriages together. Fig. 1.1 shows the arrangement.

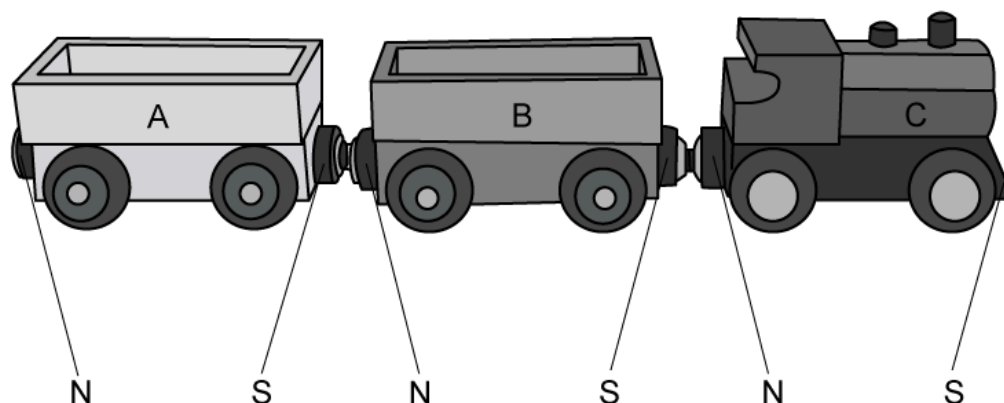


Fig. 1.1

The centre carriage (B) is removed and turned 180° .

Describe the effect this will have on the train.

(3 marks)

- (b)** Figure 1.2 shows one of the carriages with a magnet attached to each end.

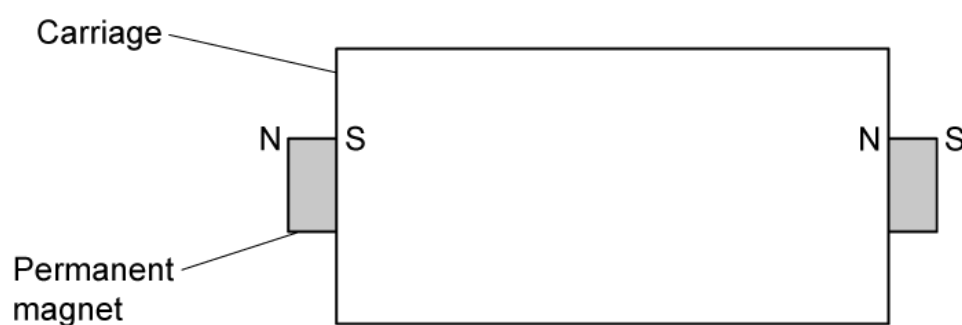


Fig. 1.2

Sketch the magnetic field lines inside the carriage.

(3 marks)

9 (a) A student experiments with electric charge.

The student uses a dry cloth to rub a plastic rod. The rod becomes positively charged.

Explain how the friction between the rod and the cloth causes the rod to become positively charged.

Use your ideas about the movement of charge.

(2 marks)

(b) The student suspends a balloon from an insulating thread, as shown in Fig. 9.1.

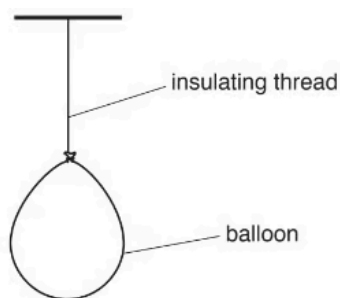


Fig. 9.1

The balloon has an electric charge.

Explain how the student can use a positively charged rod to determine the charge on the balloon.

(3 marks)

10 (a) Identify which of the following metals can be permanently magnetised. Place a tick (✓) in the box next to any correct metal.

☐ aluminium

☐ copper

☐ steel

☐ tungsten

.....
(1 mark)

(b) Two metal rods are thought to be permanent magnets. Describe the test you would carry out to confirm that both rods are permanent magnets.

.....
.....
(2 marks)

(c) (i) Describe how to make an electromagnet. You may draw a labelled diagram to help your answer.

[3]

(ii) Suggest **two** factors that affect the strength of the magnetic field of an electromagnet.

[2]

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.....
.....
.....

(5 marks)

11 (a) Radioactive sources emit α -(alpha), β -(beta) and γ -(gamma) radiations.

State which of these types of radiation can pass through paper.

(1 mark)

(b) Barium-137 is a radioactive isotope. The nuclide notation for barium-137 is



Determine the number of neutrons in a nucleus of barium-137.

number of neutrons =

(1 mark)

(c) An isotope of barium-137 has a half-life of 3 minutes.

A radioactive source contains 36 mg of this isotope.

Calculate the mass of the isotope that remains in the source after 9 minutes.

mass of the isotope remaining = mg

(3 marks)

12 (a) Explain the term *galactic redshift*.

(3 marks)

(b) Explain why galactic redshift is considered as evidence supporting the Big Bang theory.

(2 marks)